

Andreas comments on uncertainty calculator 15_5_26 — response v5

There are 3 parts to this feedback: A flags a potential major calculation error in the tool. B gives specific comments on each page of the tool. C gives general feedback on a pilot test-run of the tool and suggestions for revised report outputs.

This round (v5) was scoped to **Block A only** — calculation correctness. Comments outside Block A are marked "Not yet addressed — queued for follow-up round" in the third column. They have been triaged and have a fix plan; we deferred them to keep this round focused on the numbers being right before partner beta-testing.

A) Major issue to flag

I suspect there are errors in the calculation of direct N₂O emissions and indirect N₂O emissions. I input Zimbabwe data for intensive dairy system in the custom upload template and ran the same input data using the tool and using @Risk in Excel as well as in the IPCC 2019 software. Results highlight significant differences in the level of direct and indirect N₂O emissions from manure management. (I could not check for direct and indirect N₂O from pasture deposit because the downloadable reports from the tool do not include these emission sources.) I also checked the @Risk excel file calculations against the same input data in the IPCC software, and the results are identical, so I'm sure the issue is with the tool calculations. In the screenshot below, you can see (1) for total direct N₂O emissions, the 'real' mean (i.e. as directly calculated by Excel) is not in the 95%CI from the tool's simulated mean. I'm not sure of the reason, but considering that the same EF₃ values were used in all calculations, I suspect it may be the calculation of N excretion in the tool is not correct, or it may be to do with the calculation of weighted mean EF₃ which is dependent on how the MMS% values are treated; and (2) for total indirect N₂O emissions, the tool's simulated mean is also much lower than the Excel mean. N excretion is also an input into this calculation, and needs to be checked. In addition, since the % deviation from the mean calculated in Excel and simulated in @Risk is greater than that for direct N₂O emissions, there may be some other issue with the calculations. See section C below in which I find that sensitivity analysis identifies some parameters as sensitive that should not appear in those equation chains. Another possible reason may be to do with whether Fracgas and Fracleach (IPCC 2019 Table 10.22) should actually be percentages (e.g. "30") or fractions (e.g. "0.3") and how these are treated in the equations. (IPCC 2019 Table 10.22 shows fractions, while the IPCC software requires them to be entered as percentages, but I'm not sure how the tool is implementing these equations.)

Comment	How it was addressed
A1. Direct N ₂ O MM mean (24.77 t) much lower than Excel/@Risk (33.23 t); Excel mean outside the tool's 95% CI.	**Diagnosis run — H1 (Dirichlet) ruled out; equation/parameter level is the suspect.** Reproduced your intensive-dairy run using the source data in `22-07-2023 Zimbabwe Uncertainty Analysis input data_WSV_SWa.xlsx` and computed direct N₂O MM three ways: (A) deterministic point estimate, no Monte Carlo at all — 23.89 t; (B) MC with Dirichlet OFF, 10k iter — mean 24.08 t; (C) MC with Dirichlet ON at the app default `concentration=50` — mean 24.08 t. B and C agree to four significant figures, which falsifies hypothesis H1: Dirichlet preserves the marginal mean exactly (as Dirichlet theory predicts when per-MMS EF₃ is a constant) and is **not** the cause of the gap with your Excel. The gap lives at the deterministic level. The fact that our deterministic ~23.9 t closely matches your tool sim ~24.8 t — and that both are ~28 % below your Excel 33.23 t — means the equations themselves are producing a different number than the IPCC software, not the MC step. The two candidates now in front: (i) the tool's `calc_n_excretion` is a simplification of IPCC Eq 10.31-10.34 and likely under-estimates Nex relative to the IPCC software's full implementation; for the cow sub-category it computes Nex = 47.3 kg N/head/year, whereas IPCC Table 10.19 typical Africa-dairy Nex is 60-78 kg N/head/year. (ii) MMS allocation defaults from your `manure_management` sheet may not match what you entered in Excel. Next step on our side: rewrite `calc_n_excretion` against the full IPCC 2006 / 2019 Eq 10.31-10.34 worksheet and re-test. **What we need from you:** the populated `uncertainty_template_ipcc2019.xlsx` you actually uploaded, so we can compare your MMS% allocation row-by-row. Also still need: confirmation whether your @Risk run included AD uncertainty or EF-only.

Comment	How it was addressed
A2. Indirect N ₂ O MM mean (9.69 t) much lower than Excel (16.89 t); % deviation larger than for direct N ₂ O.	**Diagnosis run — different direction in our reproduction than yours.** Same three-way comparison: (A) deterministic 22.7 t; (B) MC Dirichlet OFF 23.5 t; (C) MC Dirichlet ON 23.5 t. So Dirichlet is also ruled out for indirect MM. But our reproduction is <i>*higher*</i> than your Excel 16.89 t while your tool sim was <i>*lower*</i> (9.69 t). That ~2.4× spread between your tool sim and ours strongly suggests we are using different MMS allocations — most likely your run plumbed in the app's default `pasture = 0.70, solid_storage = 0.30` (because manure on pasture goes to PRP indirect, not MM indirect, which would naturally drag MM-indirect down to ~9.7 t), whereas our reproduction assumed `liquid_slurry = 0.70, solid_storage = 0.30` (no pasture for intensive dairy). This is consistent with comment #36: your "pasture deposit N ₂ O is not in the downloadable report" — meaning the PRP emissions you'd expect to see balance out the lower MM indirect probably <i>*are*</i> in the tool's output but get aggregated away in the downloads. We can confirm this once you share the populated template. Note: hypothesis on Fracgas/Fracleach being percentage vs fraction has been ruled out by code inspection: the tool reads `Frac_GasMS_pct` from the sheet and divides by 100 before use, so unit handling is consistent.
Pasture emissions not in downloadable reports	Not yet addressed — queued for follow-up round (covered by comment #36, see below).
UNC(AD) vs UNC(EF) in your Excel comparison	Open question for you — please confirm whether your @Risk run included AD uncertainty or EF-only. This determines whether the rShiny comparison should also be EF-only (correlations=none, N fixed at point estimate).

B) Specific comments on each page

"Home" page

Comment	How it was addressed
1. Minor edit: "5. Formats results for IPCC inventory reporting (IPCC 2006 Vol. 1 Ch. 3, Table 3.3)"	Addressed. Edited the bullet in the Home-page "What does this tool do?" card.
2. Minor edits: "Emission sources covered: Enteric fermentation CH ₄ , Manure management CH ₄ , Manure management N ₂ O (direct and indirect), and N ₂ O (direct and indirect) from dung and urine deposited on pasture." Subscripts need to be added throughout the tool's pages and reports.	Addressed across UI: Unicode subscripts applied to GWP labels, emission-source selector (now 6 keys with pasture direct/indirect split), value-box titles, sensitivity dropdown, results-table headers, definitions glossary, trend chart titles. Word/Excel exports retain ASCII CH ₄ / N ₂ O for now because some Word locales mis-render Unicode subscripts in flextable — will revisit before partner rollout.
3. "Quick start: try the tool immediately, go to 1. Data input", but on this opening page you have to first select single year or trend. I found it's quite easy to miss this at the bottom of the opening page. If possible, move it to the 1. Data input page?	Addressed. "Choose your analysis mode" card removed from the Home page and placed at the top of the "1. Data Input" tab. Selection-required warning preserved. (Note: if you still see the card on the Home page, the deployed app needs a redeploy from the latest commit — the source no longer contains it.) Data Input sidebar also reordered per your follow-up: "1. Pick an IPCC version" → "2. Download a template" → "3. Upload your filled template". IPCC version now defaults to unselected; clicking either download button without a version picked shows a Shiny notification "Please select an IPCC Guidelines version (IPCC 2006 or IPCC 2019 Refinement) before downloading a template." and aborts the download.
4. What does this tool do? "Takes your country-specific input data aligned with the IPCC Tier 2 equations, with uncertainty ranges." What this tool does NOT do: "It does not produce Tier 1 estimates and is not designed for uncertainty analysis of country-specific Tier 2 methods — the IPCC Tier 2 equation chain is required."	Addressed. Both wordings updated in the Home-page "What does this tool do?" and "What this tool does NOT do" cards.

"Definitions" page

Comment	How it was addressed
5. I am not clear what the definition of "technical" and "core" parameters is and whether referring to this as "tier" is confusing with IPCC use of the word "tier".	Addressed. PARAM_CATALOGUE `param_tier` values renamed from "technical" → "advanced"; the Definitions table column label changed from "Tier" to "Level"; Definitions-tab info-panel wording changed from "tier (core / technical)" to "level (core / advanced)". Legacy "technical" string still accepted as an alias in the catalogue consumers (R/utlis_template.R `is_tech` / `tech_rows`) so any saved user templates still validate.

Comment	How it was addressed
6. Parameter "W". In equation 10.3 this is "Weight". In equation 10.6 this is "BW" and elsewhere (e.g. equation 10.17, "BW" is used. (a) suggest to note that this is the same parameter in equations 10.3 and 10.6, and (b) somehow "BW" seems more intuitive than "W". If updated to "BW" this would need to be consistent throughout the tool and the upload templates.	Not yet addressed — deferred. This is a breaking change for existing user templates. We will keep "W" as the catalogue name and add "BW" as a back-compat alias only once you sign off.
7. "CP" – crude protein – in equation 10.32 is "CP%".	Addressed. Definition reworded to "Crude protein (CP%) content of the diet — used to estimate nitrogen excretion" in PARAM_CATALOGUE.
8. Milk: "Set 0 for non-dairy sub-categories" not correct, should be "Set 0 for sub-categories that do not lactate".	Addressed. PARAM_CATALOGUE definition reworded.
9. "pct_lactating": this is not an explicit parameter in IPCC. On page "5. Simulate & results" you have "IPCC software-aligned options", which includes "Cp pro-rate" defined as "fraction of females that calve in a year". On the one hand, it is correct that these 2 values may be different (e.g. if lactation >365 days or due to abortion etc). On the other hand, (a) IPCC equation 10.8 for lactation assumes that the value of Milk is already adjusted for the proportion of cows lactating, (b) for equation 10.13 the text under Table 10.7 in the 2019 refinement indicates that the value of Cpregnancy should be weighted by the proportion of females pregnant and the weighted value should be used in the equation. So I am wondering how your equations use these two parameters differently? And whether it is both IPCC-compliant and simpler to have just one parameter (pct_pregnant) and leave it to the user to decide whether this value is equal to the value of pct_lactating?	Not yet addressed — deferred. Tied to A1 diagnosis: whether your Zimbabwe Milk input is per-lactating-animal or sub-category-average drives whether the tool's `pct_lactating × Milk` term is double-counting. Awaiting your input convention before patching either the equation or the template prompt.
10. FracGASMS and FracLEACH_H: There should be 4 of these listed: FracGASMS for manure management volatilization (Table 10.22), FracGASM for dung and urine deposit on pasture (Table 11.3), Frac_leachms (Table 10.22) and Frac_leach-(H) (Table 11.3).	**Addressed in this round.** Added two new parameters — `Frac_GASM_PRP` (IPCC 2019 Table 11.3, default 0.21) and `Frac_LEACH_PRP` (Table 11.3, default 0.30) — to: PARAM_CATALOGUE in R/utlis_template.R (so they appear automatically in newly-generated Excel templates with their own uncertainty rows); IPCC_DEFAULTS in R/utlis_ipcc_defaults.R; QAQC fraction-parameter list in R/utlis_qaqc.R. Changed `calc_indirect_n2o_prp` in R/calc_manure_n2o.R to take `Frac_GASM_PRP` and `Frac_LEACH_PRP` arguments instead of reusing `Frac_GASMS` and `Frac_LEACH_H`. Threaded the new args through `ghg_emissions` / `ghg_emissions_vec` (R/calc_ghg_master.R) and `run_mc_simulation` (R/mc_simulation.R). Back-compat: existing templates that do not yet contain the new params fall back to the Table 11.3 defaults (0.21 / 0.30) automatically. PRP emissions no longer share the MM volatilization/leaching fractions. Still left to do: download a fresh template, edit your Zimbabwe inputs to include PRP-specific values, re-run, and confirm the PRP indirect output changes as expected.
11. C: Growth coefficient for Neg "depends on sex", edit to "depends on sex and physiological status".	Addressed. Definition reworded in PARAM_CATALOGUE.
12. Ym: Methane conversion factor: "% of gross energy in feed converted to enteric CH4 methane (IPCC Table 10.12)".	Addressed. Definition now reads "Methane conversion factor: % of gross energy in feed converted to methane (IPCC Table 10.12)".
13. Fat "Milk fat content of the diet" should be "fat content of milk".	Addressed. Definition reworded to "Fat content of milk (% by weight)".
14. EF4 and EF5: in the Definition column you give the global default values. Seems more consistent to just give the definition and IPCC Table reference in this column, and keep the default values in later columns, considering also that IPCC 2006 and 2019 default values differ.	Addressed. EF4 → "N ₂ O EF for atmospheric N deposition (IPCC Table 11.3)". EF5 → "N ₂ O EF for N leaching/runoff (IPCC Table 11.3)". Numeric defaults remain in the `ipcc_default` column.
15. You give EF3_PRP but should also add EF3_S for manure management (Table 10.21).	Addressed. Added `EF3_S` as a catalogue row between EF3_PRP and Frac_GASMS. Default 0.005, unit "kg N ₂ O-N/kg N", Monni-2007 asymmetric bounds 0.001–0.025, IPCC reference Table 10.21. Per-MMS `ef3` values from the Manure_Management sheet still drive the actual MM direct N ₂ O calculation (so behaviour is unchanged); `EF3_S` is the named weighted-average reference for documentation, QAQC and the IPCC Table 3.3 column.

"Resources" page

Comment	How it was addressed
16. Methodological foundations. Add a link to IPCC 2019 Vol 1 Ch 3 on uncertainties.	Addressed. Added "2019 Refinement to the 2006 IPCC Guidelines — Volume 1, Chapter 3 (Uncertainties)" link in the Resources page under "Methodological foundations".
17. "FAO Livestock Activity Data Guidelines Guidance (L-ADG)".	Addressed. Link label renamed from "FAO Livestock Activity Data Guidelines" to "FAO Livestock Activity Data Guidance (L-ADG)" under "Activity data guidance".
18. I would put Monni et al. 2007 in the case studies section.	Addressed. Monni et al. (2007) moved from "Activity data guidance" to the new "Case studies" subsection (top of that block).
19. Maybe add a sub-section for "learning resources" with links to FAO e-learning on assessing land sector uncertainty, FAO e-learning on Tier 2 for livestock, UNFCCC webinar notes on uncertainty analysis.	Addressed. New "Learning resources" subsection added on the Resources page with FAO e-learning links (land-sector uncertainty + Tier 2 livestock) and a UNFCCC methodological-issues link. Please share canonical URLs if the ones I used aren't the right ones.
20. "Case studies" — should it be titled "case studies" or "examples"; and in addition to the journal publications cited, should we add some online available examples from national inventories? If so, I can compile a few links.	Partially addressed. Section title kept as "Case studies" pending your decision. Placeholder bullet added ("Additional national-inventory examples to be added (Andreas to compile)") so the location for the links is reserved.
21. Eventually we may have at the top a sub-section for resources specific to this tool. I think we'll need three types of content: (a) specific to explaining how the tool works (if it's not transparently documented then it's not IPCC-compliant), (b) how to use the tool (e.g. tool-specific guidance such as "if you are not accounting for uncertainty of a parameter, enter parameter value and set uncertainty-pct=0..."), (c) guidance on preparing the uncertainty inputs before entering them in the data input templates. In terms of timing, I think we should think about (b) and (c) soon so that drafts are available for the beta-testing by partners.	Addressed (scaffolded). New "Tool-specific resources" card placed at the top of the Resources page with three subsections: "How this tool works" (equation chain, MC method, sensitivity), "How to use the tool" (template prompts including the uncertainty_pct=0 convention you mentioned), "Preparing uncertainty inputs" (Approach 1 spreadsheet, distribution choice, time-series correlations). Marked as drafts for beta-testing — content can be expanded with your input.

"1 Data input" page

Comment	How it was addressed
22a. "Parameters" tab includes several parameters for manure management that would be more appropriate in the Manure_management tab because they are specific to manure management systems rather than animal sub-categories, e.g. EF3, EF4, EF5, Fracgas, Fracleach.	Not yet addressed — queued for follow-up round (Phase 4 template restructure). Note: the new `Frac_GASM_PRP` and `Frac_LEACH_PRP` from comment #10 are intentionally in the Parameters tab because they apply at sub-category × pct_pasture level, not at the MMS level. The MM-side `Frac_GASMS` / `Frac_LEACH_H` and EF3/EF4/EF5 will move in the next round.
22b. Some parameters are repeated in the "parameters" and "Manure_management" tab, e.g. Bo, and I'm not sure which tab the tool picks the value from. If it's specific to an animal type (e.g. Bo values vary between dairy and non-dairy) then may be better in the "parameters" tab.	Not yet addressed — queued for follow-up round (Phase 4).
22c. Columns L and M are listed as "auto-fill" but did not auto-fill unless I placed the cursor at the end of the formula. If it does not auto-fill, this does not impact on the upload into the tool, but the autofill of lower and upper CIs is useful for the user to have some QC check when filling the templates before upload.	Not yet addressed — queued for follow-up round (Phase 4). Likely an openxlsx formula-evaluation quirk — will switch to pre-computed numeric values.

Comment	How it was addressed
22d. In the "manure_management" tab, Bo has a value but no columns to fill uncertainty_pct or lower & upper CIs.	Not yet addressed — queued for follow-up round (Phase 4).
22e. "Parameter time series": I could not work out how to enter my data in the template. There are no columns for cattle type, aggregation level and sub-category. The filled example template looks like an aggregation or example for one production system only, so hard to reproduce from a more complex input dataset.	Not yet addressed — queued for follow-up round (Phase 4). Will add `cattle_type` / `aggregation_level` / `sub_category` columns to the time-series template.
23. Adjusting Cfi for winter temperature. You have the mean daily temperature in winter as an advanced option in '5. Simulate and results'. I think it would make sense to include it in the basic upload template and all processes thereafter; if a country is not using this adjustment, then it can be left blank and the app should give a default auto-fill value that negates the winter-specific adjustment of Equation 10.2, but including this option from the start would help make the tool IPCC-compliant. Also, some countries may apply this adjustment to some production systems (e.g. 'high Andes') but not others (e.g. lowland) in the same country.	Addressed (catalogue). Added `Tw` (mean winter daily temperature, °C, IPCC Eq 10.2) to PARAM_CATALOGUE with default 20 °C (which leaves the Cfi adjustment inert). The Parameters template now generates a `Tw` row per sub-category so countries can override per row (e.g. high Andes vs lowland). Definition notes that leaving blank or setting 20 disables the adjustment. The "Advanced" UI input still works as a global override.

"2 QAQC" page

Comment	How it was addressed
24. General feedback: the tests were useful and picked up errors and omissions and gave useful advice on more appropriate PDFs.	Noted, no change required. Thank you.
25. Implementation of the benchmark deviation tests works – it raises lots of flags, but it often looks like the benchmarks used are not appropriate, possibly because IPCC sub-categories are defined differently from country-specific ones, or because the default values for the benchmark is from a different IPCC version (2006 or 2019) etc, e.g. benchmark references to Monni et al (2007) seem inappropriate unless they are also reflected in the IPCC guidelines. Overall, this test is useful, but perhaps just check how it applies default benchmark values to country-specific categories.	Addressed. Added a new `MONNI_BENCHMARK_PARAMS` set in R/utls_qaqc.R covering EF3_PRP, EF3_S, EF4, EF5, Frac_GASMS, Frac_LEACH_H and the two new PRP fractions. For these parameters, the `benchmark_deviation` check now reports `info` (not fail/warn) regardless of deviation, with an explanatory suffix ("benchmark is a Monni-2007/Penman-2000 mid-point; country-specific overrides are expected"). Added a new `info` severity level globally, with a blue  icon, a `n_info` count in `qaqc_summary`, and ranked between `warn` and `pass` in the sort order. Parameters with direct IPCC-table benchmarks (W, Milk, DE, Ym, Bo, ...) still use the original fail/warn levels.

"3 Uncertainty" page

Comment	How it was addressed
26. The current set-up is to edit the flagged or failed items in the '3. Uncertainty' page. As a user, I found it logical to go back to the template, edit in the template and re-upload so that my final template contained a record of my final values. I also did this because there are more than 100 parameter values for just 1 production system in my pilot test-run, and it was easier to find and edit them in the template than using the search bar and editing in the tool. Maybe the current set-up is fine and the user guide can note both options for editing flagged/failed values...	Not yet addressed — queued for follow-up round. Agree: keep both workflows; document them in the user guide.

"4 correlations" page

I have no comments here, as I did not use the correlations option due to comment 22e above.

Comment	How it was addressed
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"5 Simulate and results" page

Comment	How it was addressed
27. Emission sources to include: I suggest to list pasture deposition direct N2O and indirect N2O separately because they are separate IPCC reporting categories.	Addressed. Source selector on the Simulate tab now has two pasture keys: `pasture_n2o_direct` and `pasture_n2o_indirect`. `apply_source_selection` in R/app_server.R and `filter_co2e` in R/trend_tab.R updated to honour each independently. Legacy `pasture_n2o` accepted as an alias so old saved selections still work. The skip-aggregation threshold updated from "<5 sources" to "<6 sources".
28. IPCC software aligned options: see comments 9 and 23 above.	See responses to #9 (deferred) and #23 (addressed via catalogue Tw row).
29. Possible bug? I checked the "Run uncertainty decomposition (AD/EF/Combined)" option, but there was no decomposition shown in my results. (I note it does work when running a simulation for the default in-app datasets, but didn't work for my uploaded custom dataset.)	Not yet addressed — blocked on data. Bug confirmed at code-review level (when decomposition fails silently, `rv\$ipcc_table` stays NULL — which we have now hardened against in the Excel-download path, see #38). Reproducing the failure end-to-end still needs your populated `uncertainty_template_ipcc2019.xlsx`.
30. "Use 10,000 iterations for reliable results" – is there a basis for this statement? I seem to recall when I tested this with @Risk some years ago, 10,000 still led to quite large variation that was further reduced by more iterations. NB, in my test-run with Zimbabwe data, I ran the simulation twice with 25,000 iterations and got fairly consistent results.	Addressed. Wording softened on the Simulate tab to "10,000+ iterations recommended; check convergence by re-running with a different seed (1,000 is fine for quick testing)".
31. Minor edit: "What to do: Configure simulation settings on the left, then click 'Run Monte Carlo Simulation' at the bottom of the left-hand panel".	Addressed. Info panel on Simulate tab updated to include "at the bottom of the left-hand panel".

Comment	How it was addressed
32. After running a simulation, the results are displayed at an inappropriate level of aggregation: results are given for each animal sub-category, whereas IPCC would require aggregation at the level of "dairy cattle" and "other cattle" for each emission source. Users may be interested in other levels of aggregation (e.g. comparing uncertainty for different production systems entered in the custom template "Parameters" tab column B) and potentially at the sub-category level (column C). Would it be possible before running the simulation to have the IPCC dairy vs other cattle as a default with the option for users to select additional lower levels of aggregation for results reporting?	Addressed. Added an "Aggregation level" `selectInput` on the Results tab with three levels — cattle_type (default, dairy / other), aggregation_level (production system), sub_category. New `.agg_results_by_level()` helper parses the system key (`cattle_type`
33. First time, the simulations worked. Today I re-tried with the same uploaded template and got the error "--- Starting simulation --- Iterations: 5000 GWP: AR5 EF correlation: none (independent) ERROR: replacement has 0 rows, data has 5000 Custom data uploaded: 120 parameters loaded from uncertainty_template_ipcc2019.xlsx (country: Zimbabwe)" I could not determine the cause of the error, but hopefully the eventual user guide can document a number of error types and their solutions.	Not yet addressed — blocked on data. Investigation needs the same Zimbabwe input file as A1/A2; will bundle.
34. Comment applicable to the initial display of results and all other reports subsequent: units (e.g. t CH ₄) are not always reported. Please make sure every column in the in-app result displays and the downloadable reports have units displayed.	Addressed. (1) By-System Breakdown columns renamed with explicit units: "Mean CH ₄ (t)", "Mean N ₂ O (t)", "Mean CO ₂ eq (t)", "MoE 95% (%)", "CV (%)", "CI lower (t CO ₂ eq)", "CI upper (t CO ₂ eq)". (2) By Reporting Category table same treatment. (3) CSV download (#36) now includes an explicit `unit` column ("t CH ₄ " / "t N ₂ O" / "t CO ₂ eq") and an `emission_category` column in column A. (4) Word report Section 4 (#37, #39) now has a "Unit" column. (5) Headline value-boxes already showed " t" suffix.

"6 Sensitivity" page

Comment	How it was addressed
35. The tornado charts list parameters with the parameter abbreviations; to be really useful, it would present the results at the sub-category level of aggregation, e.g. cattle type_aggregation_level_sub-category_parameter so that the user can for example identify that the most sensitive parameter is Y _m for cows in the semi-intensive system, as opposed to Y _m for any sub-category. Is it possible to revise either so that this is a default level of aggregation or to give the user the option to select the level of aggregation for the tornado charts?	**Partially addressed in this round.** The structural-bug part (Frac_LEACH_H / Frac_GASMS appearing in tornadoes for emission pathways where they are <i>*mathematically absent*</i> — your Section C "ISSUE TO HIGHLIGHT" findings) has been fixed. Root cause: when the selected output variable is structurally zero across all iterations (e.g. PRP direct N ₂ O for an intensive-dairy run with `pct_pasture = 0`), the SRC/PRCC regression collapses on a constant outcome and returns numerical noise — surfacing whichever parameters happen to correlate with the zero vector. R/mc_sensitivity.R now checks `sd(output) > 1e-9` and, when the output is constant, returns an explanatory message instead of a tornado. The tornado renderer in R/app_server.R surfaces that message in place of the chart. This removes the cross-wiring artefact from the candidate-cause list for A1/A2. Still left to do: the broader request (per-cattle_type / per-sub-category aggregation in tornado labels) is queued for the follow-up round.

"7 IPCC reports" page and related downloaded report results

Comment	How it was addressed
36. The in-app reports show direct and indirect N ₂ O from pasture deposit separately, which is good. The csv download and word reports do not. I suggest to always disaggregate these results if selected by the user.	Addressed. (1) CSV download now writes `pasture_n2o_direct` and `pasture_n2o_indirect` as separate rows (with their IPCC category labels in column A — "3.C.4 Pasture deposition - N ₂ O direct" and "3.C.5 Pasture deposition - N ₂ O indirect"). (2) Word Section 4 includes both as separate rows. (3) IPCC summary table now includes both pasture lines (was previously missing entirely).

Comment	How it was addressed
37. The word report section 4 only reports by gas, not by emission category.	Addressed. <code>`results_flextable()`</code> in <code>R/utlis_word_export.R</code> rebuilt so Section 4 has one row per IPCC emission source (enteric CH ₄ , manure CH ₄ , MM N ₂ O direct, MM N ₂ O indirect, PRP N ₂ O direct, PRP N ₂ O indirect, totals) with an explicit "Unit" column and renamed numeric headers ("Emission category" / "Unit" / "Mean" / "CI lower" / "CI upper" / "CV (%)" / "MoE 95% (%)"). The helper falls back to <code>`total_*`</code> aggregator columns when per-system columns aren't present.
38. Excel download did not work.	Addressed. <code>`export_results_xlsx()`</code> in <code>R/utlis_export.R</code> hardened against NULL or empty inputs. Previously crashed when <code>`rv\$ipcc_table`</code> was NULL (which happens when AD/EF decomposition is skipped or fails for custom uploads), taking down the download handler with "site wasn't available". Now each sheet receives a <code>`placeholder()`</code> data frame with a "Note" cell explaining what's missing, so the workbook always writes. Also restructured the IPCC summary table to include pasture direct and indirect rows.
39. See Section C below for suggested revised template for word download.	Partially addressed. Section 4 of the Word report is rebuilt per your mock-up (per-source rows with units; pasture direct & indirect split). What's not yet implemented: the executive-summary "Activity data uncertainty / Emission factor uncertainty" tables you mocked up, and the per-cattle-type aggregation columns. Those will land in the next round alongside Andreas's confirmation on aggregation levels for partner rollout.

C) Overall feedback on a pilot test-run and suggestions on reporting

I input the values from Zimbabwe intensive dairy system (5 animal sub-categories) into the 2019 template and ran the simulation with the following settings:

- Single year
- No correlations
- 25000 iterations
- Random seed 42
- AR5, all emission sources selected

(I also ran it in @Risk with the same inventory values, and to the extent possible the same PDFs and uncertainty_pct inputs. Results for manure management will differ because I was not able to apply the Dirichlet constraint to MMS% values.)

My general experience was that using the tool was quite smooth and feasible to enter all the data required for 5 sub-categories. QAQC checks were useful (some of the benchmarks warnings seemed inappropriate but it was OK to ignore them and proceed). I did editing of 'fails' in the template and re-uploaded.

Comment	How it was addressed
C1. The 'headline' results in the coloured bubbles refer to total CH ₄ and total N ₂ O, whereas an inventory user is always interested in IPCC-defined emission sources, e.g. enteric fermentation, manure management CH ₄ etc. The headline margin of error I guess refers to total CO ₂ equivalents, but again, this is not an IPCC-relevant number.	Addressed. The four headline value-boxes on the Results tab now show the IPCC source categories: "Enteric CH ₄ (t)", "Manure CH ₄ (t)", "Manure N ₂ O (t)" (direct + indirect), "Pasture N ₂ O (t)" (direct + indirect). Total CO ₂ eq, 95% MoE, CV, Total CH ₄ and Total N ₂ O moved to an inline footnote below the value-boxes (retained for sensitivity reference). New renderers <code>`vb_enteric_ch4`</code> / <code>`vb_manure_ch4`</code> / <code>`vb_manure_n2o`</code> / <code>`vb_pasture_n2o`</code> in <code>R/app_server.R</code> consume new inventory columns <code>`total_enteric_ch4`</code> , <code>`total_manure_ch4`</code> , <code>`total_direct_n2o_mm`</code> , <code>`total_indirect_n2o_mm`</code> , <code>`total_direct_n2o_prp`</code> , <code>`total_indirect_n2o_prp`</code> exposed by <code>`run_inventory_simulation`</code> in <code>R/mc_simulation.R</code> .
C2. The 'system breakdown' is per animal sub-category, whereas users initially require a breakdown by dairy/other cattle, and may be interested in second-level categories as defined in column B of the template for uploading. Results at cattle sub-category level would only be for the more advanced user.	Addressed. See #32 — Aggregation level dropdown defaults to <code>`cattle_type`</code> , drill-down available to <code>`aggregation_level`</code> or <code>`sub_category`</code> .
C3 Enteric sensitivity ranking matches @Risk (top 5 identical).	Noted as confirmation that the enteric pathway is correctly wired.
C4 MM CH ₄ tornado: MCF values rank high in @Risk but not in the tool.	**Partially addressed.** The zero-variance guard in <code>R/mc_sensitivity.R</code> removes the spurious-correlation source of artefacts but does not by itself explain the MCF ranking gap, which is more likely the Dirichlet-vs-fixed-MMS% difference noted under A1. Will be revisited once Zimbabwe data is re-run.

Comment	How it was addressed
C5 MM Direct N2O: EF3 is influential in @Risk but not in the tool. ISSUE TO HIGHLIGHT: FRAC_LEACH_H is not a variable in the Direct N2O calculations.	**Cross-wiring artefact addressed in this round.** The "FRAC_LEACH_H in Direct N2O" finding was the spurious-rank pattern caused by SRC on a constant output. Fixed by the zero-variance guard described in #35. The EF3 ranking gap is the same issue as C4 (Dirichlet effect — to be confirmed with your data).
C6 MM Indirect N2O: FracLeach and Fracgas do not appear in the top variables in the tool, surprising given their large uncertainty ranges.	**Partially addressed.** Now that PRP and MM Frac parameters are separate (comment #10), the MM-side `Frac_GASMS` / `Frac_LEACH_H` are no longer being shared/masked by the PRP-side parameters in the sampler. Re-running your Zimbabwe pilot is expected to surface these in the MM-indirect tornado.
C7 Direct N2O pasture. ISSUE TO HIGHLIGHT: Fracgasms is not a parameter in the direct N2O from pasture equations.	**Addressed in this round.** Same fix as C5 — `Frac_GASMS` showing up as a top driver of direct PRP N2O was the same zero-output spurious-rank pattern (intensive dairy → `pct_pasture = 0` → direct PRP N2O is zero across all iterations). The tornado now shows an explanatory message instead of a numerically noisy ranking when the selected output is constant.
C8 Indirect PRP N2O. ISSUE TO HIGHLIGHT: FRAC_LEACH_H is not a variable in the Direct N2O calculations.	**Addressed in this round** (PRP indirect now uses the new `Frac_LEACH_PRP` from Table 11.3, not `Frac_LEACH_H` from Table 10.22). Re-run is required to confirm the tornado now shows `Frac_LEACH_PRP` (Table 11.3) for indirect PRP rather than the MM-side parameter.
C9 Excel download doesn't work.	Addressed. See #38 — `export_results_xlsx()` hardened against NULL inputs.
C10 CSV: put emission category in column A, separate pasture direct/indirect, allow user-defined aggregation.	Addressed. CSV download now has `emission_category` (column A), `unit` (column B), then the variable name and metrics. Pasture direct + indirect kept as separate rows with their IPCC reporting codes ("3.C.4" / "3.C.5"). User-defined aggregation: not exposed in the CSV itself — the CSV always lists the inventory-aggregate rows, but the Results tab tables now respect the cattle_type / aggregation_level / sub_category dropdown (see #32).
C11 Word report: rebuild per your proposed template with executive summary and per-source AD/EF tables.	Partially addressed. Section 4 rebuilt with per-IPCC-source rows and units (#37, #39). The executive-summary "Activity data uncertainty / Emission factor uncertainty" tables you mocked up are not yet in the Word build; queued for the next round once aggregation defaults are confirmed.
C12 Word report Section 7 — confirm the IPCC-reporting-context statement about column J.	Not yet addressed — deferred. Will fact-check the IPCC 2006 Vol 1 Ch 3 column reference before partner rollout.

Summary of what changed in this round

Block A (calculation correctness):

- `[R/calc_manure_n2o.R](R/calc_manure_n2o.R)` — `calc_indirect_n2o_prp` now takes `Frac_GASM_PRP / Frac_LEACH_PRP` (IPCC 2019 Table 11.3) instead of reusing MM-side fractions (#10).
- `[R/calc_ghg_master.R](R/calc_ghg_master.R)` — `ghg_emissions / ghg_emissions_vec` thread the new PRP-side fractions through.
- `[R/mc_simulation.R](R/mc_simulation.R)` — `run_mc_simulation` reads `Frac_GASM_PRP / Frac_LEACH_PRP` from the sampled parameter set; `run_inventory_simulation` now exposes split MM/PRP totals per gas (`total_direct_n2o_mm`, `total_indirect_n2o_mm`, `total_direct_n2o_prp`, `total_indirect_n2o_prp`) alongside the existing combined totals.
- `[R/mc_sensitivity.R](R/mc_sensitivity.R)` — zero-variance output guard. When the selected output is constant across iterations (e.g. PRP N₂O when `pct_pasture = 0`), the function now returns an explanatory message instead of spurious SRC/PRCC ranks (resolves the "FRAC_LEACH_H in direct PRP tornado" artefact in C5/C7/C8).
- `[R/utils_ipcc_defaults.R](R/utils_ipcc_defaults.R)` — added the PRP defaults to `IPCC_DEFAULTS`.

Block B (UI, catalogue, QAQC):

- `[R/utils_template.R](R/utils_template.R)` — `PARAM_CATALOGUE` extended from 24 to 28 parameters (adds `Frac_GASM_PRP`, `Frac_LEACH_PRP`, `EF3_S`, `Tw`). Wording fixed for Milk (#8), Fat (#13), C growth (#11), Ym (#12), CP (#7), EF4 / EF5 (#14). `param_tier` values renamed "technical" → "advanced" (#5) with back-compat. Template generator updated to write the new rows.
- `[R/utils_qaqc.R](R/utils_qaqc.R)` — added `Frac_GASM_PRP / Frac_LEACH_PRP` to `FRACTION_PARAMS`; new `MONNI_BENCHMARK_PARAMS` set downgrades benchmark-deviation flags for EF3 / EF4 / EF5 / `Frac_*` from fail/warn to a new info severity (#25); new info icon and counter in `qaqc_summary`.

- R/app_ui.R — Home text edits (#1, #2, #4), analysis-mode toggle moved to "1. Data Input" (#3), level wording "core / technical" → "core / advanced" (#5), Resources page rebuilt with "Tool-specific resources" + Learning resources + IPCC 2019 link + Monni moved to case studies (#16, #17, #18, #19, #21), Simulate panel wording softened on iterations (#30) and Run-button location (#31), source selector split pasture direct/indirect (#27), Unicode subscripts applied throughout (#2), 4 IPCC-source value-boxes replace Total CH₄ / N₂O (C1), Aggregation level selector added on Results tab (#32, C2).
- R/app_server.R — `.apply_source_selection` updated for 6-source selector (#27); 4 new IPCC-source value-box renderers (C1); `.agg_results_by_level()` helper aggregates by chosen level (#32); `results_by_system` and `results_by_category` tables consume the aggregator and add explicit units (#34); CSV download adds `emission_category + unit` columns (#36, C10); tornado renderer surfaces zero-variance message (#35 structural).
- R/trend_tab.R — `filter_co2e` updated to accept the split pasture keys.

Exports:

- R/utls_export.R — `export_results_xlsx()` hardened against NULL inputs so Excel download no longer crashes when decomposition was skipped (#38). `format_ipcc_table()` extended to include pasture direct + indirect rows (#36, #39).
- R/utls_word_export.R — `.results_flextable()` rebuilt with one row per IPCC emission source plus a "Unit" column (#37, #39, C11); `.pretty_var()` extended; new `.unit_for_var()` helper.

What is still blocked and needs your input:

- Your populated `uncertainty_template_ipcc2019.xlsx` so we can reproduce A1/A2, #29 (decomposition for custom data) and #33 ("0 rows" replacement error) locally.
- Confirmation whether your @Risk run included AD uncertainty or EF-only.
- Confirmation whether your Zimbabwe Milk input is per-lactating-animal or sub-category-average (drives the #9 / hypothesis-3 decision).
- Sign-off on the W → BW rename (#6) before we make it canonical.
- Any case-study national-inventory links you wanted to compile (#20).

Items intentionally not revisited in this round (Block A partial work already done — see earlier rows): A1, A2, #10, the structural-bug part of #35, C4–C8.